

Ex. No: 14 File Allocation Strategies

PROGRAM A: SEQUENTIAL FILE ALLOCATION

C PROGRAM

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
int start, length, i;
```

```
printf("Enter Starting Block: ");
```

```
scanf("%d", &start);
```

```
printf("Enter File Length (Number of Blocks): ");
```

```
scanf("%d", &length);
```

```
printf("\nAllocated Blocks:\n");
```

```
for(i = 0; i < length; i++)
```

```
{
```

```
printf("%d ", start + i);
```

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```
}
```

```
printf("\n");
```

```
return 0;
```

```
}
```

Output:

```
(kali㉿kali)-[~]  
$ nano seq.c  
  
(kali㉿kali)-[~]  
$ gcc seq.c -o seq  
  
(kali㉿kali)-[~]  
$ ./seq  
Enter Register Number: 2  
Enter Name: pasupathi  
  
Record Details  
Register Number : 2  
Name : pasupathi
```

SHELL SCRIPT

```
#!/bin/bash  
  
echo "Enter Starting Block:"  
read start  
  
echo "Enter File Length:"  
read length  
  
echo "Allocated Blocks:"  
for ((i=0;i<length;i++))  
do  
echo -n "${(start+i)} "  
done  
echo
```

Output:

```

(kali㉿kali)-[~]
$ nano lab14-1.sh

(kali㉿kali)-[~]
$ chmod +x lab14-1.sh

(kali㉿kali)-[~]
$ bash lab14-1.sh
Enter Starting Block:
10
Enter File Length:
5
Allocated Blocks:
10 11 12 13 14

```

PROGRAM B: INDEXED FILE ALLOCATION

C PROGRAM

```

#include <stdio.h>

int main()
{
    int n, indexBlock, blocks[20], i;
    printf("Enter Index Block: ");
    scanf("%d", &indexBlock);
    printf("Enter Number of Blocks: ");
    scanf("%d", &n);
    printf("Enter Block Numbers:\n");

```

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```

for(i = 0; i < n; i++)
{

```

```

scanf("%d", &blocks[i]);
}

printf("\nIndex Block : %d\n", indexBlock);

printf("Allocated Blocks : ");

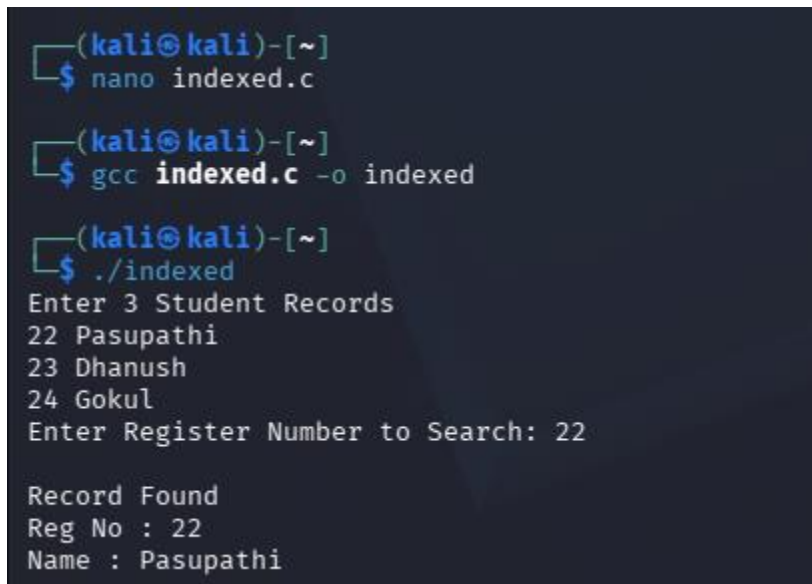
for(i = 0; i < n; i++)
{
printf("%d ", blocks[i]);
}

printf("\n");

return 0;
}

```

Output:



```

(kali@kali)-[~]
$ nano indexed.c

(kali@kali)-[~]
$ gcc indexed.c -o indexed

(kali@kali)-[~]
$ ./indexed
Enter 3 Student Records
22 Pasupathi
23 Dhanush
24 Gokul
Enter Register Number to Search: 22

Record Found
Reg No : 22
Name : Pasupathi

```

SHELL SCRIPT

```
#!/bin/bash
```

```
echo "Enter Index Block:"
```

```
read index
echo "Enter Number of Blocks:"
read n
echo "Enter Block Numbers:"
for ((i=0;i<n;i++))
do
read block[$i]
done
echo "Index Block : $index"
echo -n "Allocated Blocks : "
for ((i=0;i<n;i++))
do
echo -n "${block[$i]} "
done
echo
```

Output:

```

(kali㉿kali)-[~]
$ nano lab14-2.sh

(kali㉿kali)-[~]
$ chmod +x lab14-2.sh

(kali㉿kali)-[~]
$ bash lab14-2.sh
Enter Index Block:
5
Enter Number of Blocks:
4
Enter Block Numbers:
10
20
30
40
Index Block : 5
Allocated Blocks : 10 20 30 40

```

PROGRAM C: LINKED FILE ALLOCATION

C PROGRAM

```

#include <stdio.h>

int main()
{
    int n, blocks[20], i;

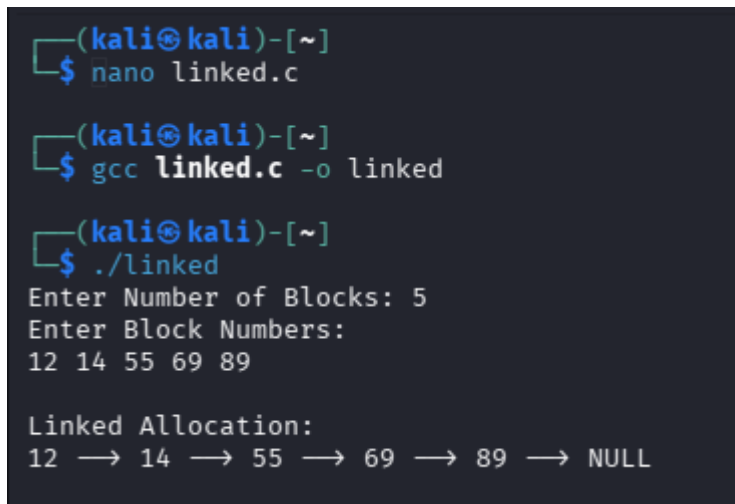
    printf("Enter Number of Blocks: ");
    scanf("%d", &n);
    printf("Enter Block Numbers:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &blocks[i]);
    }

    printf("\nLinked Allocation:\n");
    for(i = 0; i < n - 1; i++)
    {

```

```
printf("%d --> ", blocks[i]);  
}  
printf("%d --> NULL\n", blocks[n - 1]);  
return 0;  
}
```

Output:

A terminal window with a dark background and light blue/green text. The prompt is (kali㉿kali)-[~]. The user enters 'nano linked.c'. The prompt is (kali㉿kali)-[~]. The user enters 'gcc linked.c -o linked'. The prompt is (kali㉿kali)-[~]. The user enters './linked'. The program outputs: 'Enter Number of Blocks: 5', 'Enter Block Numbers:', '12 14 55 69 89', 'Linked Allocation:', and '12 → 14 → 55 → 69 → 89 → NULL'.

```
(kali㉿kali)-[~]  
$ nano linked.c  
  
(kali㉿kali)-[~]  
$ gcc linked.c -o linked  
  
(kali㉿kali)-[~]  
$ ./linked  
Enter Number of Blocks: 5  
Enter Block Numbers:  
12 14 55 69 89  
  
Linked Allocation:  
12 → 14 → 55 → 69 → 89 → NULL
```

SHELL SCRIPT

```
#!/bin/bash  
  
echo "Enter Number of Blocks:"  
read n  
  
echo "Enter Block Numbers:"  
for ((i=0;i<n;i++))  
do  
read block[$i]  
done  
  
echo "Linked Allocation:"  
for ((i=0;i<n-1;i++))  
do
```

```
echo -n "${block[$i]} --> "
```

```
done
```

```
echo "${block[$((n-1))]} --> NULL"
```

Output:

```
(kali㉿kali)-[~]  
$ nano lab14-3.sh  
  
(kali㉿kali)-[~]  
$ chmod +x lab14-3.sh  
  
(kali㉿kali)-[~]  
$ bash lab14-3.sh  
Enter Number of Blocks:  
4  
Enter Block Numbers:  
5  
12  
34  
56  
Linked Allocation:  
5 → 12 → 34 → 56 → NULL
```